Homework #7

7.1 Assume that W/L ratios of Figure 7.1 are $(W/L)_1 = 2\mu m/1\mu m$ and $(W/L)_2 = (W/L)_3 = (W/L)_4 = 1\mu m/1\mu m$. Find the dc value of v_{IN} that will give a dc current in M1 of 110μA. Calculate the small signal voltage gain and output resistance using the parameters of Table 7.1. Assume $\lambda=\gamma=0$.

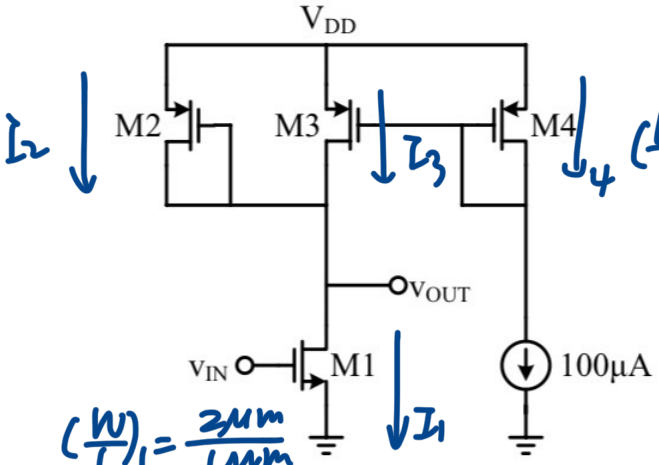


Figure 7.1

I_2 ↓ I_3 ↓ I_4 ↓ $(\frac{W}{L})_2=(\frac{W}{L})_3=(\frac{W}{L})_4=\frac{1\mu m}{1\mu m}$

$K=\mu_n C_{ox}=134$

$K_p=\mu_p C_{ox}=50$

$V_{TH}=0.7V$

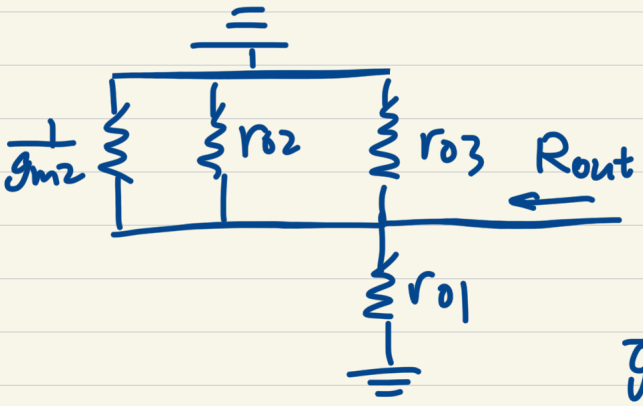
$\lambda_{pmos}=0.2V^{-1}$

$\lambda_{nmos}=0.1V^{-1}$

解. $I_1=\frac{1}{2}\mu_n C_{ox}(\frac{W}{L})_1(V_{in}-V_{TH})^2$

$\Rightarrow V_{in}=\sqrt{\frac{2I_1}{\mu_n C_{ox}(\frac{W}{L})_1}}+V_{TH}=1.606V$

$I_3=I_4=100\mu A$, $I_1=110\mu A \Rightarrow I_2=10\mu A$



$r_{O1}=\frac{1}{\lambda I_1}=0.0909M\Omega$

$r_{O2}=\frac{1}{\lambda I_2}=0.5M\Omega$

$r_{O3}=\frac{1}{\lambda I_3}=0.05M\Omega$ } $0.0303M\Omega$

$\frac{1}{g_{m2}}=\frac{1}{\sqrt{2\mu_n C_{ox}(\frac{W}{L})_2 I_2}}=0.0316M\Omega$

故 $R_{out}=r_{O1}\,||\,r_{O2}\,||\,r_{O3}\,||\,\frac{1}{g_{m2}}=15.47K\Omega$

$A_v=-g_{m1}R_{out}=-\sqrt{2\mu_n C_{ox}(\frac{W}{L})_1 I_1}\,R_{out}=-3.76$

若 $\lambda=0$, 即 $r_{O1}, r_{O2}, r_{O3}\to\infty$, 则 $r_{out}=\frac{1}{g_{m2}}=31.6K\Omega$

$A_v=-g_{m1}R_{out}=-\sqrt{2\mu_n C_{ox}(\frac{W}{L})_1 I_1}\,R_{out}=-7.67$

7.2 Suppose the common-source stage of Fig 7.2 is to provide an output swing from 1V to 2.5V. Assume that $(W/L)_1 = 50/0.5$, $R_D = 2k\Omega$, $V_{DD} = 3V$ and $\lambda = 0$. Use model parameters in Table 7.1.

a) Calculate the input voltages that yield $V_{out} = 1V$ and $V_{out} = 2.5V$.

b) Calculate the drain current and the transconductance of M_1 for both cases.

c) How much does the small-signal gain, g_mR_D, vary as the output goes from 1V to 2.5V?

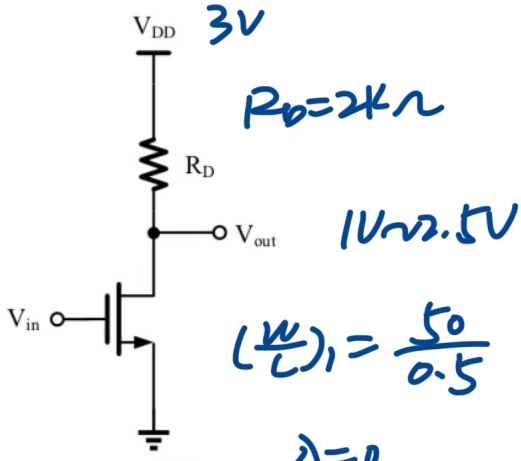


Figure 7.2

解. (a). $V_{out}=1V$ 时, $I=\frac{V_{DD}-V_{out}}{R_D}=1mA$

假设饱和, $I=\frac{1}{2}\mu_nC_{ox}(\frac{W}{L})_1(V_{GS}-V_{TH})^2$

$\Rightarrow V_{in}=\sqrt{\frac{2I}{\mu_nC_{ox}(\frac{W}{L})_1}}+V_{TH}=1.036V$

$V_{out}>V_{in}-V_{TH}$, 验证在饱和区

$V_{out}=2.5V$ 时, $I=\frac{V_{DD}-V_{out}}{R_D}=250\mu A$

$\Rightarrow V_{in}=\sqrt{\frac{2I}{\mu_nC_{ox}(\frac{W}{L})_1}}+V_{TH}=0.893V$

$V_{out}>V_{in}-V_{TH}$, 验证在饱和区

(b) drain current (a)已得

$g_m=\sqrt{2\mu_nC_{ox}(\frac{W}{L})I}=5.176mS\ /\ 2.588mS$

(c) $|A_v|=g_mR_D=10.352\ /\ 5.176$

$|A_v|: 5.176\sim10.352 \quad \Delta|A_v|=5.176$

7.3 Consider the circuit of Fig 7.3 with $(W/L)_1 = 50/0.5$ and $(W/L)_2 = 10/0.5$. Assume that $\lambda = \gamma = 0$, $V_{DD} = 3V$

a) At what input voltage is M_1 at the edge of the triode region? What is the small-signal gain under this condition?

b) When V_{out} is 0.66 V, what is the small-signal gain under this condition?

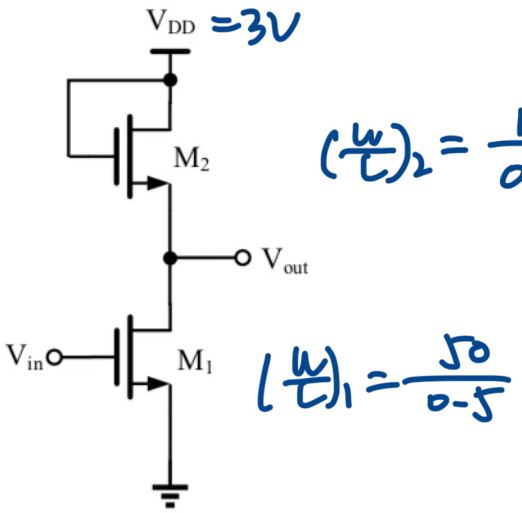


Figure 7.3

解. (a). $V_{out} = V_{in} - V_{TH}$

$I_1 = \frac{1}{2}\mu_n C_{ox}(\frac{W}{L})_1 (V_{in} - V_{TH})^2$

$I_2 = \frac{1}{2}\mu_n C_{ox}(\frac{W}{L})_2 (V_{DD} - V_{out} - V_{TH})^2 = \frac{1}{2}\mu_n C_{ox}(\frac{W}{L})_2 (V_{DD} - V_{in})^2$

$1 = \frac{I_1}{I_2} = \frac{(W/L)_1}{(W/L)_2}\left(\frac{V_{in} - V_{TH}}{V_{DD} - V_{in}}\right)^2$

令 $K = \sqrt{\frac{(W/L)_2}{(W/L)_1}}$ 则 $V_{in} = \frac{\sqrt{K}V_{DD} + V_{TH}}{1 + \sqrt{K}} = 1.414V$

$g_m = \mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{TH}) \Rightarrow g_{m1} = 9.583mS,\ g_{m2} = 4.250mS$

$A_v = -g_{m1}\frac{1}{g_{m2}} = -2.255$

(b) $I_2 = \frac{1}{2}\mu_n C_{ox}(\frac{W}{L})_2 (V_{DD} - V_{out} - V_{TH})^2 = 3.604mA$

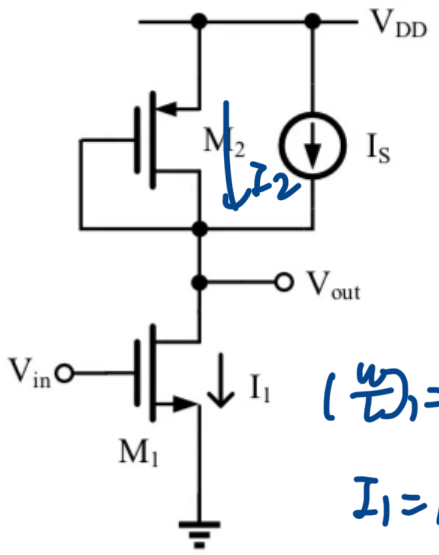
$g_{m2} = \mu_n C_{ox}(\frac{W}{L})_2 (V_{DD} - V_{out} - V_{TH}) = 4.395mS$

$V_{out} < 0.714V$, 线性区, $I_1 = \mu_n C_{ox}(\frac{W}{L})_1 [(V_{GS} - V_{TH})V_{DS} - \frac{1}{2}V_{DS}^2]$

$g_{m1} = \frac{\partial I_1}{\partial V_{GS}} = \mu_n C_{ox}(\frac{W}{L})_1 V_{DS} = 8.244mS$

$A_v = -g_{m1}\frac{1}{g_{m2}} = -2.012$

7.4 In the circuit of Fig 7.4, $(W/L)_1 = 20/0.5$, $I_1 = 1mA$, and $I_S = 0.75mA$. Assuming $\lambda = 0$, $V_{DD} = 3V$, calculate $(W/L)_2$ such that M_1 is at the edge of triode region. What is the small-signal voltage gain under this condition? Use model parameters in Table 7.1.



$V_{DD} = 3V$

$I_S = 0.75mA$

$\lambda = 0$

$(\frac{W}{L})_1 = \frac{20}{0.5}$

$I_1 = 1mA$

Figure 7.4

解:

$$I_1 = \frac{1}{2}\mu_n C_{ox}(\frac{W}{L})_1 (V_{in} - V_{TH})^2$$

$$\Rightarrow V_{out} = V_{in} - V_{TH} = \sqrt{\frac{2I_1}{\mu_n C_{ox}(\frac{W}{L})_1}} = 0.611V$$

$$I_2 = I_1 - I_S = 0.25mA$$

$$I_2 = \frac{1}{2}\mu_p C_{ox}(\frac{W}{L})_2 (V_{DD} - V_{out} - V_{THp})^2$$

$$\Rightarrow (\frac{W}{L})_2 = \frac{2I_2}{\mu_p C_{ox}(V_{DD} - V_{out} - V_{THp})^2} = 3.961$$

$$g_{m1} = \sqrt{2\mu_n C_{ox}(\frac{W}{L})_1 I_1} = 3.274mS$$

$$g_{m2} = \sqrt{2\mu_p C_{ox}(\frac{W}{L})_2 I_2} = 0.315mS$$

$$A_v = -g_{m1}\frac{1}{g_{m2}} = -10.39$$

7.5 Consider the circuit of Fig 7.5 with $(W/L)_1 = 50/0.5$, $R_D = 2k\Omega$, and $R_S = 200\ \Omega$, $V_{DD} = 3V$. Use model parameters in Table 7.1.

a) Calculate the small-signal voltage gain if $I_D = 0.5mA$.

b) Assuming that $\lambda = \gamma = 0$, calculate the input voltage that places M1 at the edge of the triode region. What is the gain under this condition?

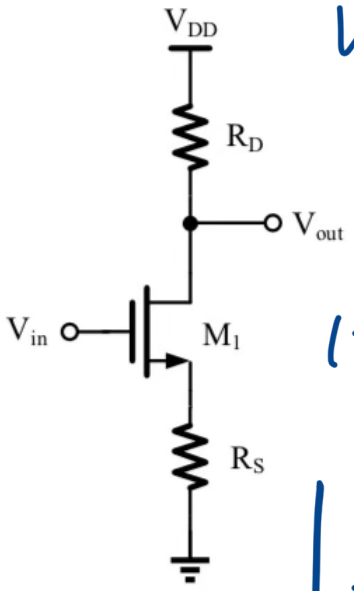


Figure 7.5

$V_{DD}=3V$

$R_D=2k\Omega$

$(\frac{W}{L})_1=\frac{50}{0.5}$

$R_S=0.2k\Omega$

$I_D=0.5mA$

$K=134\mu A/V^2$

$\lambda=0.1V^{-1}$

解. (a) $V_{out}=V_{DD}-I_DR_D=2V$, $V_S=I_DR_S=0.1V$

$V_{TH}=V_{TH0}+\gamma(\sqrt{|2\phi_F+V_{SB}|}-\sqrt{|2\phi_F|})=0.723V$

$g_m=\sqrt{2\mu_nC_{ox}(\frac{W}{L})_1 I_D(1+\lambda V_{DS})}=3.993mS$

$A_V=-\frac{R_D}{\frac{1}{g_m}+R_S}=-4.440$

(b)

$V_{out}=V_{DD}-I_DR_D$ ①

$V_{out}=V_{in}-V_{TH}$ ②

$V_S=I_DR_S$

$I_D=\frac{1}{2}\mu_nC_{ox}(\frac{W}{L})_1(V_{in}-V_S-V_{TH})^2$

由①, ② $V_{in}=V_{DD}+V_{TH}-I_DR_D$

故 $I_D=\frac{1}{2}\mu_nC_{ox}(\frac{W}{L})_1(V_{DD}-I_DR_D-I_DR_S)^2$

$$I_D = 6.7\times10^{-3}(3-2200I_D)^2$$

$$32428I_D^2 - 89.44I_D + 0.0603 = 0$$

$$\Rightarrow I_D = 1.535\,mA \text{ 或 } I_D = 1.173\,mA$$

① $I_D = 1.535\,mA$ 时，$V_{in} = 0.53V$，$V_S = 0.317V$

$$V_{GS} = 0.213V < V_{TH}$$，舍去

② $I_D = 1.173\,mA$ 时，$V_{in} = 1.354V$，$V_S = 0.2346V$

$$V_{GS} = 1.1194V > V_{TH}$$

$$g_m = \mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{TH}) = 5.620mS$$

$$A_V = -\frac{R_D}{\frac{1}{g_m}+R_S} = -5.292$$